

Module – 3

8051 Timers and Serial Port

SYLLABUS:

Timers/Counters & Serial port programming: Basics of Timers & Counters, Data types & Time delay in the 8051 using C, Programming 8051 Timers, Mode 1 & Mode 2 Programming, Counter Programming (Assembly Language only). (Text book 2- 3.4, Text book 1- 7.1, 9.1,9.2)

Basics of Serial Communication, 8051 Connection to RS232, Programming the 8051 to transfer data serially & to receive data serially using C. (Text book 2- 3.5, Text book 1- 10.1,10.2,10.3 except assembly language programs, 10.5)

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TEXT, REFERENCE & ADDITIONAL REFERENCE BOOKS

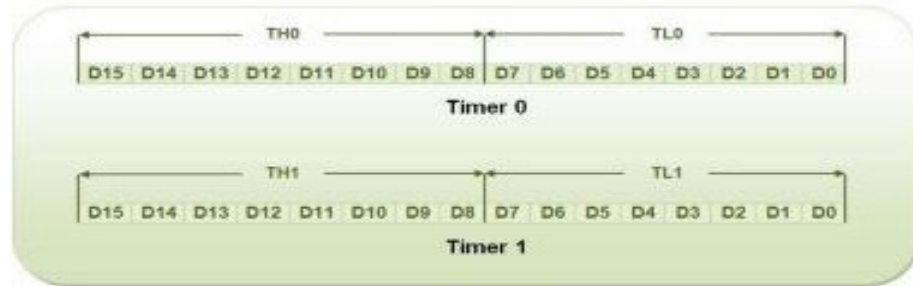
BOOK TITLE/AUTHORS/PUBLICATION/LINK	
T-1	“The 8051 Microcontroller and Embedded Systems - Using Assembly and C”, Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollind. Mckinlay; Phi, 2006 / Pearson, 2006.
T-1	“The 8051 Microcontroller”, Kenneth j. Ayala, 3rd edition, Thomson/Cengage Learning.
T-1	“Programming and Customizing The 8051 Microcontroller”, Myke Predko, Tata Mc Graw-Hill Edition 1999 (reprint 2003).
R-1	“The 8051 Microcontroller Based Embedded System”, Manish K Patel, McGraw Hill, 2014, ISBN: 978-93-329-0125-4.
R-2	Microcontrollers: Architecture, Programming, Interfacing and System Design”, Raj Kamal, Pearson Education, 2005

3.1 TIMERS:

Timers/Counters are used generally for

- Time reference
- Creating delay
- Wave form properties measurement
- Periodic interrupt generation
- The 8051 has two timers/counters, they can be used either as Timers to generate a time delay or as Event counters to count events happening outside the microcontroller
- 8051 has two timers, Timer 0 and Timer 1.
- Timer 0 and Timer 1 are 16 bits.

- 8051 has an 8-bit architecture, each 16-bits timer is accessed as two separate registers of low byte and high byte.
- The low byte register is called TL0/TL1 and the high byte register is called TH0/TH1.
- Accessed like any other register.



Timer in 8051 is used as timer, counter and baud rate generator. Timer always counts up irrespective of whether it is used as timer, counter, or baud rate generator: Timer is always incremented by the microcontroller. The time taken to count one digit up is based on master clock frequency.

If Master CLK=12 MHz,

Timer Clock frequency = Master CLK/12 = 1 MHz

Clock Period = 1 micro second

This indicates that one increment in count will take 1 microsecond.

The two timers in 8051 share two SFRs (TMOD and TCON) which control the timers, and each timer also has two SFRs dedicated solely to itself (TH0/TL0 and TH1/TL1).

The following are timer related SFRs in 8051.

SFR Name	Description	SFR Address
TH0	Timer 0 High Byte	8Ch
TL0	Timer 0 Low Byte	8Ah
TH1	Timer 1 High Byte	8Dh
TL1	Timer 1 Low Byte	8Bh
TCON	Timer Control	88h
TMOD	Timer Mode	89h

3.1.1 TMOD Register

- Both timers 0 and 1 use the same register, called TMOD (timer mode), to set the various timer operation modes
- TMOD is an 8-bit register
- The lower 4 bits are for Timer 0, the upper 4 bits are for Timer 1

- In each case, the lower 2 bits are used to set the timer mode, the upper 2 bits to specify the operation.

GATE	C/T	M1	M0	GATE	C/T	M1	M0
TIMER 1				TIMER 0			

GATE When TRx (in TCON) is set and GATE = 1, TIMER/COUNTERx will run only while INTx pin is high (hardware control). When GATE = 0, TIMER/COUNTERx will run only while TRx = 1 (software control).

C/T Timer or Counter selector. Cleared for Timer operation (input from internal system clock). Set for Counter operation (input from Tx input pin).

M1 Mode selector bit (NOTE 1).

M0 Mode selector bit (NOTE 1).

Note 1 :

M1	M0	OPERATING MODE	
0	0	0	13-bit Timer
0	1	1	16-bit Timer/Counter
1	0	2	8-bit Auto-Reload Timer/Counter
1	1	3	(Timer 0) TL0 is an 8-bit Timer/Counter controlled by the standard Timer 0 control bits, TH0 is an 8-bit Timer and is controlled by Timer 1 control bits.
1	1	3	(Timer 1) Timer/Counter 1 stopped.

3.1.2 TCON (timer control) register

- TCON (timer control) register is an 8-bit register

TCON : Timer/Counter Control Register (Bit Addressable)

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0
-----	-----	-----	-----	-----	-----	-----	-----

TF1	TCON.7	Timer 1 overflow flag. Set by hardware when the Timer/Counter 1 overflows. Cleared by hardware as processor vectors to the interrupt service routine.
TR1	TCON.6	Timer 1 run control bit. Set/cleared by software to turn Timer/Counter ON/OFF.
TF0	TCON.5	Timer 0 overflow flag. Set by hardware when the Timer/Counter 0 overflows. Cleared by hardware as processor vectors to the service routine.
TR0	TCON.4	Timer 0 run control bit. Set/cleared by software to turn Timer/Counter 0 ON/OFF.
IE1	TCON.3	External Interrupt 1 edge flag. Set by hardware when External interrupt edge is detected. Cleared by hardware when interrupt is processed.
IT1	TCON.2	Interrupt 1 type control bit. Set/cleared by software to specify falling edge/low level triggered External Interrupt.
IE0	TCON.1	External Interrupt 0 edge flag. Set by hardware when External Interrupt edge detected. Cleared by hardware when interrupt is processed.
IT0	TCON.0	Interrupt 0 type control bit. Set/cleared by software to specify falling edge/low level triggered External Interrupt.

3.2 TIMER MODES

Timers can operate in four different modes. They are as follows

3.2.1 Timer Mode-0:

In this mode, the timer is used as a 13-bit UP counter as follows.

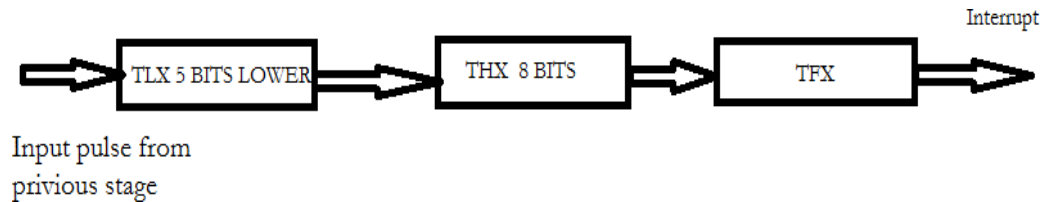


Fig. Operation of Timer on Mode-0

The lower 5 bits of TLX and 8 bits of THX are used for the 13-bit count. Upper 3 bits of TLX are ignored. When the counter rolls over from all 0's to all 1's, TFX flag is set and an interrupt is generated. The input pulse is obtained from the previous stage. If TR1/0 bit is 1 and Gate bit is 0, the counter continues counting up. If TR1/0 bit is 1 and Gate bit is 1, then the operation of the counter is controlled by input. This mode is useful to measure the width of a given pulse fed to input.

3.2.2 Timer Mode-1:

This mode is similar to mode-0 except for the fact that the Timer operates in 16-bit mode.

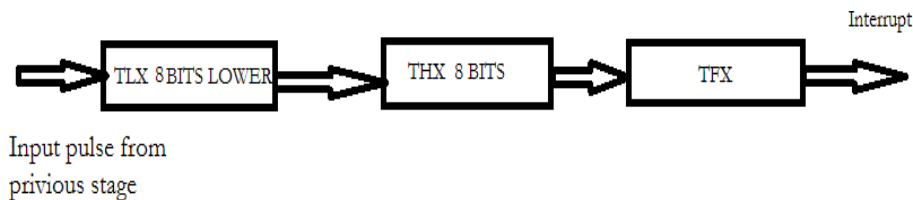


Fig: Operation of Timer in Mode 1

3.2.3 Timer Mode-2: (Auto-Reload Mode):

This is an 8-bit counter/timer operation. Counting is performed in TLX while THX stores a constant value. In this mode when the timer overflows i.e. TLX becomes FFH, it is fed with the value stored in THX. For example, if we load THX with 50H then the timer in mode 2 will count from 50H to FFH. After that 50H is again reloaded. This mode is useful in applications like fixed time sampling.

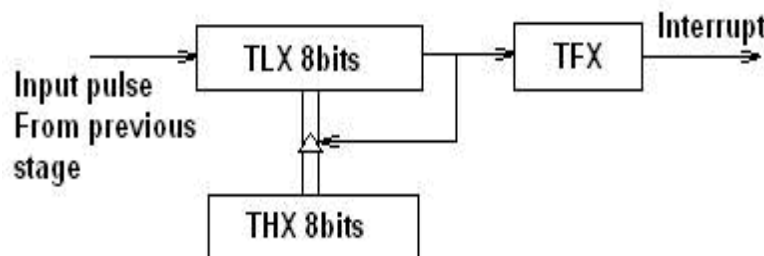


Fig: Operation of Timer in Mode 2

3.2.4 Timer Mode-3:

Timer 1 in mode-3 simply holds its count. The effect is same as setting TR1=0. Timer 0 in mode-3 establishes TL0 and TH0 as two separate counters.

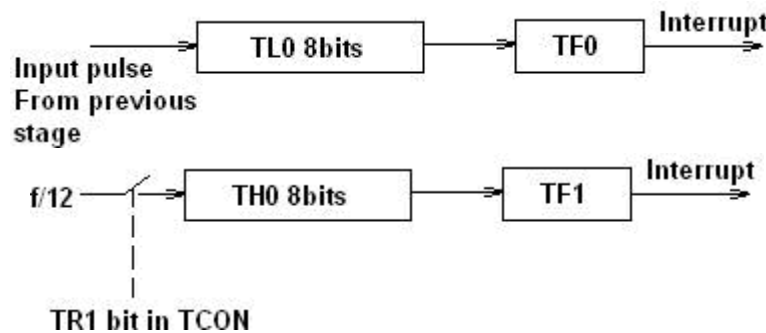


Fig: Operation of Timer in Mode 3

Control bits TR1 and TF1 are used by Timer-0 (higher 8 bits) (TH0) in Mode-3 while TR0 and TF0 are available to Timer-0 lower 8 bits (TL0).

3.3 PROGRAMMING 8051 TIMERS IN ASSEMBLY

In order to program 8051 timers, it is important to know the calculation of initial count value to be stored in the timer register. The calculations are as follows.

In any mode, Timer Clock period = $1/\text{Timer Clock Frequency}$.
 $= 1/(\text{Master Clock Frequency}/12)$

Mode 1 (16 bit timer/counter)

Value to be loaded in decimal = $65536 - (\text{Delay required}/\text{Timer clock period})$

Convert the answer into hexadecimal and load onto THx and TLx register. ($65536D = FFFFH+1$)

Mode 0 (13 bit timer/counter)

Value to be loaded in decimal = $8192 - (\text{Delay required}/\text{Timer clock period})$ Convert the answer into hexadecimal and load onto THx and TLx register. ($8192D = 1FFFH+1$)

Mode 2 (8 bit auto reload)

Value to be loaded in decimal = $256 - (\text{Delay required}/\text{Timer clock period})$

Convert the answer into hexadecimal and load onto THx register. Upon starting the timer this value from THx will be reloaded to TLx register. ($256D = FFH+1$)

3.3.1 Steps for programming timers in 8051

Mode 1:

- Load the TMOD value register indicating which timer (0 or 1) is to be used and which timer mode is selected.
- Load registers TL and TH with initial count values. Start the timer by the instruction “SETB TR0” for timer 0 and “SETB TR1” for timer 1.

- Keep monitoring the timer flag (TF) with the “JNB TFx, target” instruction to see if it is raised. Get out of the loop when TF becomes high.
- Stop the timer with the instructions “CLR TR0” or “CLR TR1”, for timer 0 and timer 1, respectively.
- Clear the TF flag for the next round with the instruction “CLR TF0” or “CLR TF1”, for timer 0 and timer 1, respectively.
- Go back to step 2 to load TH and TL again.

Mode 0:

The programming techniques mentioned here are also applicable to counter/timer mode 0. The only difference is in the number of bits of the initialization value.

Mode 2:

- Load the TMOD value register indicating which timer (0 or 1) is to be used; select timer mode 2.
- Load TH register with the initial count value. As it is an 8-bit timer, the valid range is from 00 to FFH.
- Start the timer.
- Keep monitoring the timer flag (TFx) with the “JNB TFx, target” instruction to see if it is raised. Get out of the loop when TFx goes high.
- Clear the TFx flag.
- Go back to step 4, since mode 2 is auto-reload.

Example 1

Indicate which mode and which timer are selected for each of the following.

(a) MOV TMOD, #01H (b) MOV TMOD, #20H (c) MOV TMOD, #12H

Solution:

We convert the value from hex to binary. we have:

- (a) TMOD = 00000001, mode 1 of timer 0 is selected.
 (b) TMOD = 00100000, mode 2 of timer 1 is selected.
 (c) TMOD = 00010010, mode 2 of timer 0, and mode 1 of timer 1 are selected

Example 2

Find the timer's clock frequency and its period for various 8051-based system, with the crystal frequency 11.0592 MHz when C/T bit of TMOD is 0.



$$1/12 \times 11.0592 \text{ MHz} = 921.6 \text{ KHz}; T = 1/921.6 \text{ kHz} = 1.085 \text{ us}$$

Example 3

In the following program, we create a square wave of 50% duty cycle (with equal portions high and low) on the P1.5 bit. Timer 0 is used to generate the time delay. Analyze the program

```

HERE:    MOV TMOD,#01    ;Timer 0, mode 1(16-bit mode)
         MOV TL0,#0F2H   ;TL0=F2H, the low byte
         MOV TH0,#0FFH   ;TH0=FFH, the high byte
         CPL P1.5        ;toggle P1.5
         ACALL DELAY
         SJMP HERE

DELAY:   SETB TR0        ;start the timer 0
AGAIN:   JNB TF0,AGAIN   ;monitor timer flag 0 ;until it rolls over
         CLR TR0         ;stop timer 0
         CLR TF0         ;clear timer 0 flag
         RET

```

In the above program notice the following step.

1. TMOD is loaded.
2. FFF2H is loaded into TH0-TL0.
3. P1.5 is toggled for the high and low portions of the pulse.
4. The DELAY subroutine using the timer is called.
5. In the DELAY subroutine, timer 0 is started by the SETB TR0 instruction.
6. Timer 0 counts up with the passing of each clock, which is provided by the crystal oscillator. As the timer counts up, it goes through the states of FFF3, FFF4, FFF5, FFF6, FFF7, FFF8, FFF9, FFFA, FFFB, and so on until it reaches FFFFH. One more clock rolls it to 0, raising the timer flag(TF0=1). At that point, the JNB instruction fails through.



7. Timer 0 is stopped by the instruction CLR TR0. The DELAY subroutine ends, and the process is repeated. Notice that to repeat the process, we must reload the TL and TH registers, and start the process is repeated

Example 4

In the above example, calculate the amount of time delay in the DELAY subroutine generated by the timer. Assume XTAL = 11.0592 MHz.

Solution:

The timer works with a clock frequency of 1/12 of the XTAL frequency; therefore, we have $11.0592 \text{ MHz} / 12 = 921.6 \text{ kHz}$ as the timer frequency. As a result, each clock has a period of T

$=1/921.6\text{kHz} = 1.085\mu\text{s}$. In other words, Timer 0 counts up each $1.085\mu\text{s}$ resulting in delay = number of counts $\times 1.085\mu\text{s}$.

The number of counts for the roll over is $\text{FFFFH} - \text{FFF2H} = 0\text{DH}$ (13decimal). However, we add one to 13 because of the extra clock needed when it rolls over from FFFF to 0 and raise the TF flag. This gives $14 \times 1.085\mu\text{s} = 15.19\mu\text{s}$ for half the pulse. For the entire period it is $T = 2 \times 15.19\mu\text{s} = 30.38\mu\text{s}$ as the time delay generated by the timer.

Example 5

The following program generates a square wave on P1.5 continuously using timer 1 for a time delay. Find the frequency of the square wave if $\text{XTAL} = 11.0592\text{ MHz}$. In your calculation do not include the overhead due to Instructions in the loop.

```

AGAIN:      MOV  TMOD,#10    ;Timer 1, mod 1 (16-bitmode)
            MOV  TL1,#34H    ;TL1=34H, low byte of timer
            MOV  TH1,#76H    ;TH1=76H, high byte timer
            SETB TR1         ;start the timer 1 BACK:
            JNB  TF1,BACK     ;till timer rolls over
            CLR  TR1         ;stop the timer 1
            CPL  P1.5        ;comp. p1. to get hi, lo
            CLR  TF1         ;clear timer flag 1
            SJMP AGAIN       ;is not auto-reload

```

Solution:

Since $\text{FFFFH} - 7634\text{H} = 89\text{CBH} + 1 = 89\text{CCH}$ and $89\text{CCH} = 35276$ clock count and $35276 \times 1.085\mu\text{s} = 38.274\text{ ms}$ for half of the square wave. The frequency = 13.064Hz . Also notice that the high portion and low portion of the square wave pulse are equal. In the above calculation, the overhead due to all the instruction in the loop is not included.

Example 6

Write a program to continuously generate a square wave of 2 kHz frequency on pin P1.5 using timer 1. Assume the crystal oscillator frequency to be 12 MHz .

The period of the square wave is $T = 1/(2\text{ kHz}) = 500\mu\text{s}$. Each half pulse = $250\mu\text{s}$.

The value n for $250\mu\text{s}$ is: $250\mu\text{s} / 1\mu\text{s} = 250$ $65536 - 250 = \text{FF06H}$. $\text{TL} = 06\text{H}$ and $\text{TH} = 0\text{FFH}$.

```

AGAIN:      MOV  TMOD,#10    ;Timer 1, mode 1
            MOV  TL1,#06H    ;TL0 = 06H
            MOV  TH1,#0FFH   ;TH0 = FFH
            SETB TR1         ;Start timer 1
BACK:       JNB  TF1,BACK     ;Stay until timer rolls over
            CLR  TR1         ;Stop timer 1
            CPL  P1.5        ;Complement P1.5 to get Hi, Lo
            CLR  TF1         ;Clear timer flag 1
            SJMP AGAIN       ;Reload timer

```

Example 7

Write a program segment that uses timer 1 in mode 2 to toggle P1.0 once whenever the counter reaches a count of 100. Assume the timer clock is taken from external source P3.5 (T1).

The TMOD value is 60H. The initialization value to be loaded into TH1 is $256 - 100 = 156 = 9CH$

```

MOV TMOD, #60h ;Counter1, mode 2, C/T'= 1MOV
TH1, #9Ch      ;Counting 100 pulses
SETB P3.5      ;Make T1 input
SETB TR1       ;Start timer 1
BACK: JNB TF1, BACK ;Keep doing it if TF = 0CPL
      P1.0      ;Toggle port bit
CLR TF1        ;Clear timer overflow flag
SJMP BACK      ;Keep doing it

```

3.4 DATA TYPES AND TIME DELAY IN THE 8051 USING C

A good understanding of C data types for 8051 can help programmers to create smaller hex files

- Unsigned char
- Signed char
- Unsigned int
- Signed int
- Sbit (single bit)
- Bit and sfr

3.4.1 Unsigned char:

- The character data type is the most natural choice as 8051 is an 8-bit microcontroller
- Unsigned char is an 8-bit data type in the range of 0 – 255 (00 – FFH)
- One of the most widely used data types for the 8051
- C compilers use the signed char as the default if we do not put the keyword unsigned

Example 8

Write an 8051 C program to send values 00 – FF to port P1.

Solution:

```

#include <reg51.h>
void main(void)
{
unsigned char z;
for (z=0; z<=255; z++)
P1=z;
}

```

3.4.2 Signed char:

Signed char is an 8 bit data type in which the MSB represents the sign (+ or -) in the range - 128 to +127

3.4.3 Unsigned int

Unsigned int is a 16 bit data type which takes the value in the range 0 to 65535 (0 – FFFF). This is used for:

- Defining 16 bit memory addresses
- To set the counter values which is more than 256

3.4.4 Signed int

Signed int is a 16 bit data type. MSB is the sign bit and the remaining 15 bits represent the magnitude from - 32768 to +32767

3.4.5 Sbit (single bit)

This data type is used in case of accessing a single bit addressable register. It allows to access to the single bit SFR registers

3.4.6 Bit and sfr

Bit data type is used for accessing the bit addressable memory of RAM (20h-2fh).

Sfr data type is used for accessing a SFR register by another name. All the SFR registers must be declared with capital letters

Example 9

Write an 8051 C program to toggle all the bits of P1 continuously.

Solution:

```
//Toggle P1 forever
#include <reg51.h>
void main(void)
{
for (;;)
{
p1=0x55;
p1=0xAA;
}
}
```

3.5 COUNTER PROGRAMMING

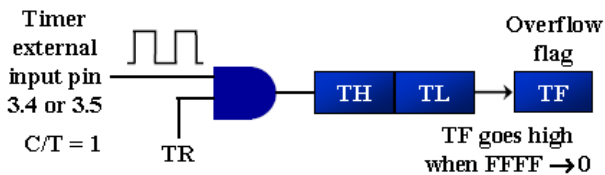
Timers can also be used as counters, counting events happening outside the 8051

- When it is used as a counter, it is a pulse outside of the 8051 that increments the TH, TL registers
- TMOD and TH, TL registers are the same as for the timer discussed previously
- The C/T bit in the TMOD registers decides the source of the clock for the timer
- When $C/T = 1$, the timer is used as a counter and gets its pulses from outside the 8051
- The counter counts up as pulses are fed from pins 14 and 15, these pins are called T0 (timer 0 input) and T1 (timer 1 input)

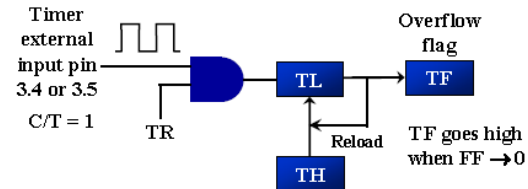
Port 3 pins used for Timers 0 and 1

Pin	Port Pin	Function	Description
14	P3.4	T0	Timer/counter 0 external input
15	P3.5	T1	Timer/counter 1 external input

Timer with external input (Mode 1)



Timer with external input (Mode 2)



Example 10

Assuming that clock pulses are fed into pin T1, write a program for counter 1 in mode 2 to count the pulses and display the state of the TL1 count on P2, which connects to 8 LEDs.

Solution:

```

MOV TM0D,#01100000B    ;counter 1, mode 2, C/T=1 external pulses
MOV TH1,#0              ;clear TH1
SETB P3.5               ;make T1 input
AGAIN: SETB TR1          ;start the counter
BACK: MOV A,TL1          ;get copy of TL
MOV P2,A                ;display it on port 2
JNB TF1,Back             ;keep doing, if TF = 0
CLR TR1                 ;stop the counter 1
CLR TF1                 ;make TF=0
SJMP AGAIN               ;keep doing it

```

Notice in the above program the role of the instruction SETB P3.5. Since ports are set up for output when the 8051 is powered up, we make P3.5 an input port by making it high. In other words, we must configure (set high) the T1 pin (pin P3.5) to allow pulses to be fed into it.

3.6 SERIAL COMMUNICATION

The 8051 microcontroller is parallel device that transfers eight bits of data simultaneously over eight data lines to parallel I/O devices. Parallel data transfer over a long is very expensive. Hence, a serial communication is widely used in long distance communication. In serial data communication, 8-bit data is converted to serial bits using a parallel in serial out shift register and then it is transmitted over a single data line. The data byte is always transmitted with least significant bit first.

3.7 BASICS OF SERIAL DATA COMMUNICATION

3.7.1 Communication Links

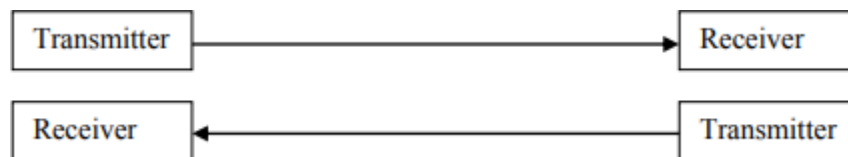
1. **Simplex communication link:** In simplex transmission, the line is dedicated for transmission. The transmitter sends and the receiver receives the data.



2. **Half duplex communication link:** In half duplex, the communication link can be used for either transmission or reception. Data is transmitted in only one direction at a time.



3. **Full duplex communication link:** If the data is transmitted in both ways at the same time, it is a full duplex i.e. transmission and reception can proceed simultaneously. This communication link requires two wires for data, one for transmission and one for reception.

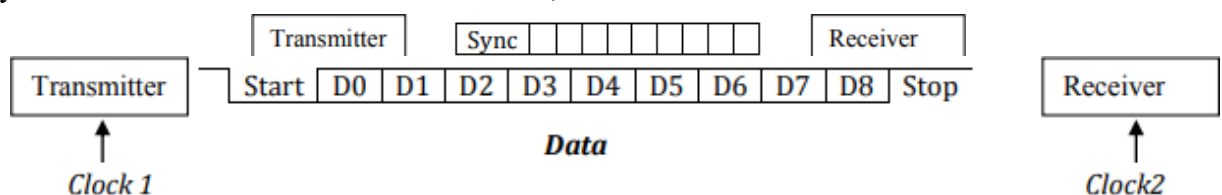


3.7.2 Types of Serial communication:

Serial data communication uses two types of communication.

1. **Synchronous serial data communication:** In this transmitter and receiver are synchronized. It uses a common clock to synchronize the receiver and the transmitter. First the synch character is sent and then the data is transmitted. This format is generally used for high speed transmission. In Synchronous serial data communication, a block of data is transmitted at a time.

2. **Asynchronous Serial data transmission:** In this, different clock sources are used for transmitter



and receiver. In this mode, data is transmitted with start and stop bits. A transmission begins with start bit, followed by data and then stop bit. For error checking purpose parity bit is included just prior to stop bit. In Asynchronous serial data communication, a single byte is transmitted at a time.

3.7.3 Baud rate:

The rate at which the data is transmitted is called baud or transfer rate. The baud rate is the reciprocal of the time to send one bit. In asynchronous transmission, baud rate is not equal to number of bits per second. This is because; each byte is preceded by a start bit and followed by parity and stop bit. For example, in synchronous transmission, if data is transmitted with 9600 baud, it means that 9600 bits are transmitted in one second. For bit transmission time = $1 \text{ second} / 9600 = 0.104 \text{ ms}$.

3.8 8051 SERIAL COMMUNICATION

Three special function registers support serial communication.

3.8.1 SBUF Register:

Serial Buffer (SBUF) register is an 8-bit register. It has separate SBUF registers for data transmission and for data reception. For a byte of data to be transferred via the TXD line, it must be placed in SBUF register. Similarly, SBUF holds the 8-bit data received by the RXD pin and read to accept the received data.

3.8.2 SCON register:

The contents of the Serial Control (SCON) register are shown below. This register contains mode selection bits, serial port interrupt bit (TI and RI) and also the ninth data bit for transmission and reception (TB8 and RB8).

SM0	SM1	SM2	REN	TB8	RB8	TI	RI
-----	-----	-----	-----	-----	-----	----	----

SM0	SCON.7	Serial port mode specifier
SM1	SCON.6	Serial port mode specifier
SM2	SCON.5	Used for multiprocessor communication
REN	SCON.4	Set/cleared by software to enable/disable reception
TB8	SCON.3	Not widely used
RB8	SCON.2	Not widely used
TI	SCON.1	Transmit interrupt flag. Set by HW at the begin of the stop bit mode 1. And cleared by SW
RI	SCON.0	Receive interrupt flag. Set by HW at the begin of the stop bit mode 1. And cleared by SW

Note: Make SM2, TB8, and RB8 = 0

SM0, SM1: They determine the framing of data by specifying the number of bits per character, and the start and stop bits

SM0	SM1	
0	0	Serial Mode 0
0	1	Serial Mode 1, 8-bit data, 1 stop bit, 1 start bit
1	0	Serial Mode 2
1	1	Serial Mode 3

SM2: This enables the multiprocessing capability of the 8051

REN (receive enable) : It is a bit-addressable register. When it is high, it allows 8051 to receive data on RXD pin. If low, the receiver is disabled.

TI (transmit interrupt) : When 8051 finishes the transfer of 8-bit character. It raises TI flag to indicate that it is ready to transfer another byte. TI bit is raised at the beginning of the stop bit.

RI (receive interrupt) : When 8051 receives data serially via RXD, it gets rid of the start and stop bits and places the byte in SBUF register. It raises the RI flag bit to indicate that a byte has been received and should be picked up before it is lost. RI is raised halfway through the stop bit.

3.8.3 PCON register:

PCON : Power Control Register (Not Bit Addressable)

SMOD	-	-	-	GF1	GF0	PD	IDL
------	---	---	---	-----	-----	----	-----

SMOD	PCON.7	Double baud rate bit. If SMOD = 1, the baud rate is doubled when the serial port is used in mode 1, 2 and 3.
-	PCON.6	Not implemented, reserved for future use*
-	PCON.5	Not implemented, reserved for future use*
-	PCON.4	Not implemented, reserved for future use*
GF1	PCON.3	General purpose bit.
GF0	PCON.2	General purpose bit.
PD	PCON.1	Power Down bit. If set, the oscillator is stopped. A reset or an interrupt (80C154 and 80C154D only) can cancel this mode (Note 1).
IDL	PCON.0	IDLE bit. If set the activity CPU is stopped. A reset or an interrupt can cancel this mode (See Note 1).

- 1.
- 2.

The SMOD bit (bit 7) of PCON register controls the baud rate in asynchronous mode transmission.

3.9 SERIAL COMMUNICATION MODES

3.9.1 Mode 0:

In this mode serial port runs in synchronous mode. The data is transmitted and received through RXD pin and TXD is used for clock output. In this mode the baud rate is 1/12 of clock frequency.

3.9.2 Mode 1:

In this mode SBUF becomes a 10 bit full duplex transceiver. The ten bits are 1 start bit, 8 data bit and 1 stop bit. The interrupt flag TI/RI will be set once transmission or reception is over. In this mode the baud rate is variable and is determined by the timer 1 overflow rate.

$$\begin{aligned}\text{Baudrate} &= [2^{\text{smod}}/32] \times \text{Timer 1 overflow Rate} \\ &= [2^{\text{smod}}/32] \times [\text{Oscillator Clock Frequency}] / [12 \times [256 - [\text{TH1}]]]\end{aligned}$$

3.9.3 Mode 2:

This is similar to mode 1 except 11 bits are transmitted or received. The 11 bits are, 1 start bit, 8 data bit, a programmable 9th data bit, 1 stop bit.

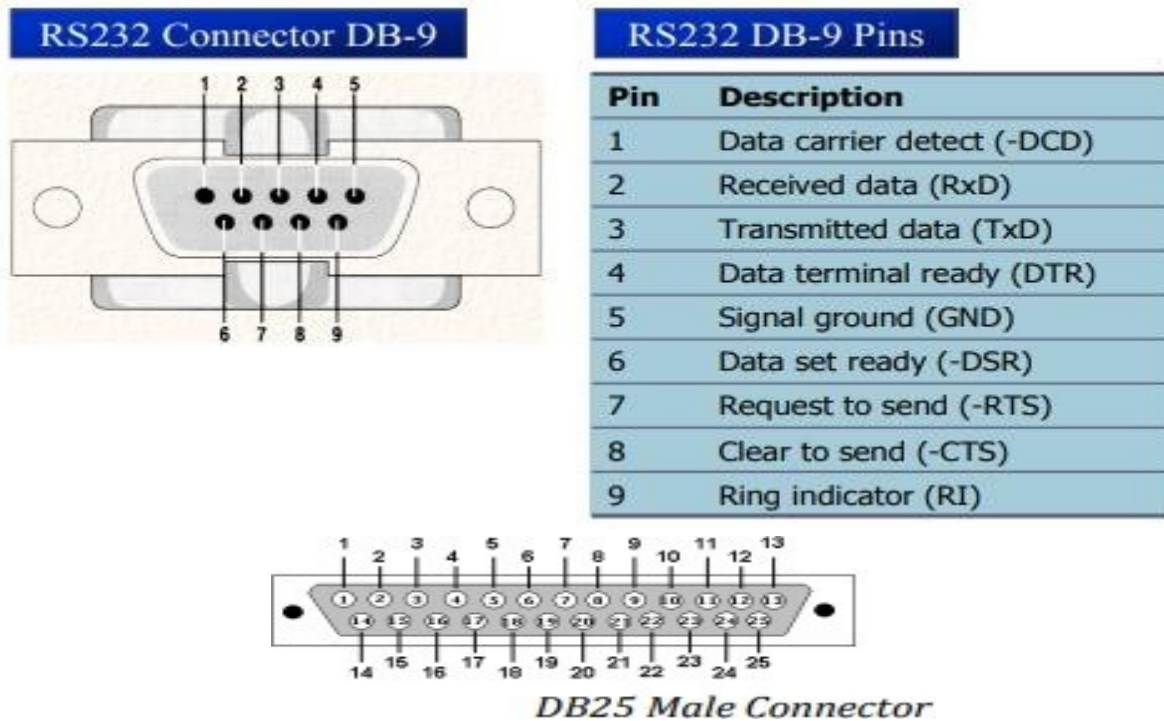
$$\text{Baud rate} = [2^{\text{smod}}/64] \times \text{Oscillator Clock Frequency}.$$

3.9.4 Mode 3:

This is similar to mode 2 except baud rate is calculated as in mode 1

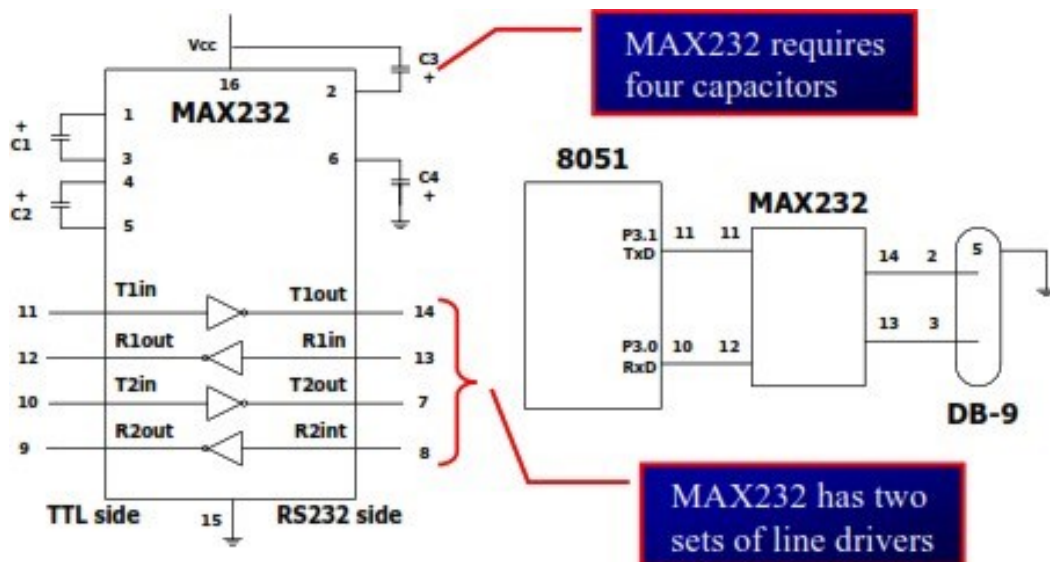
3.10 RS-232 standards:

To allow compatibility among data communication equipment made by various manufactures, an interfacing standard called RS232 was set by the Electronics Industries Association (EIA) in 1960. Since the standard was set long before the advent of logic family, its input and output voltage levels are not TTL compatible. In RS232, a logic one (1) is represented by -3 to -2.5V and referred as MARK while logic zero (0) is represented by +3 to +2.5V and referred as SPACE. For this reason, to connect any RS232 to a microcontroller system we must use voltage converters such as MAX232 to convert the TTL logic level to RS232 voltage levels and vice-versa. MAX232 IC chips are commonly referred as line drivers. In RS232 standard we use two types of connectors. DB9 connector or DB25 connector.



3.10.1 The 8051 connection to MAX232 is as follows.

The 8051 has two pins that are used specifically for transferring and receiving data serially. These two pins are called TXD, RXD. Pin 11 of the 8051 (P3.1) assigned to TXD and pin 10 (P3.0) is designated as RXD. These pins TTL compatible; therefore they require line driver (MAX 232) to make them RS232 compatible. MAX 232 converts RS232 voltage levels to TTL voltage levels and vice versa. One advantage of the MAX232 is that it uses a +5V power source which is the same as the source voltage for the 8051. The typical connection diagram between MAX 232 and 8051 is shown below



Programming Serial Data Transmitting

1. TMOD register is loaded with the value 20H, indicating the use of timer 1 in mode 2 (8-bit auto-reload) to set baud rate
2. The TH1 is loaded with one of the values to set baud rate for serial data transfer
3. The SCON register is loaded with the value 50H, indicating serial mode 1, where an 8-bit data is framed with start and stop bits
4. TR1 is set to 1 to start timer 1
5. TI is cleared by CLR TI instruction
6. The character byte to be transferred serially is written into SBUF register
7. The TI flag bit is monitored with the use of instruction JNB TI,xx to see if the character has been transferred completely
8. To transfer the next byte, go to step 5

Write a program for the 8051 to transfer letter "A" serially at 4800 baud, continuously.

Solution:

```
MOV    TMOD,#20H    ;timer 1,mode 2(auto reload)
MOV    TH1,#-6      ;4800 baud rate
MOV    SCON,#50H    ;8-bit, 1 stop, REN enabled
SETB   TR1          ;start timer 1
AGAIN: MOV    SBUF,#"A" ;letter "A" to transfer
HERE:  JNB    TI,HERE  ;wait for the last bit
        CLR    TI      ;clear TI for next char
        SJMP   AGAIN    ;keep sending A
```

Write a program for the 8051 to transfer “YES” serially at 9600 baud, 8-bit data, 1 stop bit, do this continuously

Solution:

```

MOV    TMOD,#20H    ;timer 1,mode 2(auto reload)
MOV    TH1,#-3      ;9600 baud rate
MOV    SCON,#50H    ;8-bit, 1 stop, REN enabled
SETB   TR1          ;start timer 1
AGAIN: MOV    A,#"Y"    ;transfer "Y"
        ACALL TRANS
        MOV    A,#"E"    ;transfer "E"
        ACALL TRANS
        MOV    A,#"S"    ;transfer "S"
        ACALL TRANS
        SJMP  AGAIN      ;keep doing it
;serial data transfer subroutine
TRANS: MOV    SBUF,A    ;load SBUF
HERE:   JNB    TI,HERE   ;wait for the last bit
        CLR    TI        ;get ready for next byte
        RET

```

Programming Serial Data Receiving

1. TMOD register is loaded with the value 20H, indicating the use of timer 1 in mode 2 (8-bit auto-reload) to set baud rate
2. TH1 is loaded to set baud rate
3. The SCON register is loaded with the value 50H, indicating serial mode 1, where an 8-bit data is framed with start and stop bits
4. TR1 is set to 1 to start timer 1
5. RI is cleared by CLR RI instruction
6. The RI flag bit is monitored with the use of instruction JNB RI,xx to see if an entire character has been received yet
7. When RI is raised, SBUF has the byte, its contents are moved into a safe place
8. To receive the next character, go to step 5

Write a program for the 8051 to receive bytes of data serially, and put them in P1, set the baud rate at 4800, 8-bit data, and 1 stop bit

Solution:

```

MOV    TMOD,#20H    ;timer 1,mode 2(auto reload)
MOV    TH1,#-6      ;4800 baud rate
MOV    SCON,#50H    ;8-bit, 1 stop, REN enabled
SETB   TR1          ;start timer 1
HERE:  JNB    RI,HERE ;wait for char to come in
MOV    A,SBUF       ;saving incoming byte in A
MOV    P1,A         ;send to port 1
CLR    RI           ;get ready to receive next
                        ;byte
SJMP   HERE         ;keep getting data

```

3.11 C PROGRAMS

Example 11

Write a C program for 8051 to transfer the letter “A” serially at 4800 baud continuously. Use 8-bit data and 1 stop bit.

Solution:

```

#include <reg51.h>
void main(void)
{
    TMOD=0x20;    //use Timer 1, mode 2
    TH1=0xFA;     //4800 baud rate
    SCON=0x50;
    TR1=1;
    while (1) {
        SBUF='A';    //place value in buffer
        while (TI==0);
        TI=0;
    }
}

```

Example 12

Write an 8051 C program to transfer the message “YES” serially at 9600 baud, 8-bit data, 1 stop bit. Do this continuously.

Solution:

```
#include <reg51.h>
void SerTx(unsigned char);
void main(void){
    TMOD=0x20; //use Timer 1, mode 2
    TH1=0xFD; //9600 baud rate
    SCON=0x50;
    TR1=1; //start timer
    while (1) {
        SerTx('Y');
        SerTx('E');
        SerTx('S');
    }
}

void SerTx(unsigned char x){
    SBUF=x; //place value in buffer
    while (TI==0); //wait until transmitted
    TI=0;
}
```

Example 13

Program the 8051 in C to receive bytes of data serially and put them in P1. Set the baud rate at 4800, 8-bit data, and 1 stop bit.

Solution:

```
#include <reg51.h>
void main(void){
    unsigned char mybyte;
    TMOD=0x20; //use Timer 1, mode 2
    TH1=0xFA; //4800 baud rate
    SCON=0x50;
    TR1=1; //start timer
    while (1) { //repeat forever
        while (RI==0); //wait to receive
        mybyte=SBUF; //save value
        P1=mybyte; //write value to port
        RI=0;
    }
}
```

3.12 QUESTION BANK

1. Explain mod 2 operations of timers and explain steps involved in programming timer in mod 2
2. Analyze the structure of TMOD register
3. Analyze the structure of TCON register
4. Explain MODE-1 programming of timers in 8051.
5. Write an ALP to generate square wave of frequency 1KHz on P1.3. Assume crystal frequency, XTAL=22MHz. Use Timer 1 in mode 1.
6. Assume XTAL = 22MHz. Write an ALP to generate a square wave of frequency 1KHz on P1.2.
7. Assuming XTAL frequency as 11.0592MHz write a program to generate 4KHz square wave on P2.1. Use timer 0 in mode 1. Show all the calculations.
8. Write the steps to program the timer of 8051 in mode 2.
9. Write a program for counter1 in mode 2 to count pulse and display the state of TL1 count on port 2. Assume that the clock is connected to pin 3.5.
10. Write an assembly level program to generate a square wave on P1.3 of frequency 2KHz. Use timer1 in mode 1 and show the delay calculation. XTAL used 11.0592MHz.
11. Write an assembly language program to generate a square wave on port pin P1.2 of frequency 5k Hz.
12. Explain how timers are used as counters, explain the counters operation using a code snippet.
13. Assume XTAL = 11.0592 MHz, write an assembly language program to generate a square wave of 50 kHz frequency on pin P2.3.
14. Explain RS232 standard and 9 pin DB connector in serial communication
15. Write a C program to transfer "YES" serially at 9600 baud rate, 8 bit data, 1 stop bit, do this continuously.
16. Explain full duplex, half duplex and simplex in serial communication
17. Write the steps required for programming 8051 to transmit and receive the data serially.
18. Explain the importance of TI flag and RI flag.
19. Write a C program for 8051 to transfer letter "A" serially at 4800 baud continuously. Use 8-bit data and 1 stop bit.
20. Write a 8051 C program to transmit the message ECE using serial communication port of 8051 use baud rate 4800.
21. Explain the bit pattern of SCON register with diagram..
22. Explain the importance of MAX232 IC with its pin details.
23. Write an assembly language program to transfer multi-byte data serially with 9600 baud rate.